

DATASET · IMPLEMENTATION FLOW · ACTUAL OUTPUTS

NeuroTrace-DAG

Separate group document for **dataset / input, phase-by-phase Input → Processing → Output**, and **measured results** from **outputs_junyi/phases/*.json** (Phases 1–6, Follow-up 3, probe_freeze, paper_* eval-validity JSONs). Numbers are loaded from JSON at build time — not hand-typed. Study framing: **evaluation-validity** (domain replication/extension of next-basket reality-check work), not a recommender win. Covers the seven required topics: (1) datasets, (2) phase I/O, (3) travel plan, (4) actual outputs/samples, (5) visualizations, (6) ROC/AUC, (7) full end-to-end flow. Readable companion: **docs/DATASET_INPUT_FLOW_RESULTS.md** and Word **docs/NeuroTrace-DAG_Dataset_Input_Flow_Results.docx**. Data cards: docs/data_cards/. Rebuild: **python scripts/build_dataflow_report.py --mode=full**. Demo: <http://127.0.0.1:8020>.

Product / diagnostic output: (i) P(review | history), (ii) ranked next exercises tagged review/advance with predicted recall, (iii) neighbourhood DAG. **Confirmatory scope (after prereg-v1)**: primary H-rev (native↔cluster GRU–Markov sign flip; family $m \leq 6$, Bonferroni) on the freeze holdout — see §4.13. RAGR, forgetting, H-rep, and DAG readiness are **exploratory** (graph/forget Junyi-only). We do **not** claim a practically superior recommender, faithful GRU4Rec/SASRec reproductions, or that we discovered repetition bias itself. Canonical capped ranking CI is **review_ranking.json**; full-test and calibration live in **paper_ranking_full.json** / **paper_calibration.json**. Prereg drafts: docs/PREREGISTRATION.md + PREREGISTRATION_REPLICATION.md + DEVIATIONS.md.

1. Dataset / input

1.1 What datasets are we using?

Primary (implementation + demo): Junyi Academy Math Practicing Log via EduData **ktbd-junyi** (Chang, Lin & Chen, EDM 2015; DataShop #1198). Same id space for exercises, prerequisite DAG, and learner attempts.

Supporting (cross-checks only): (a) ASSISTments 2009–2010 skill-builder **corrected** CSV (EduData USTC mirror) for skill-level repetition and rank-reversal; (b) XES3G5M (NeurIPS 2023) question-level sequences for KC-level repetition and rank-reversal; (c) Junyi raw timed extract (**timed_interactions.npz**) for learner-disjoint day-gap forgetting and timed ranking. OULAD×MOOCCubeX was rejected (no shared concept ids).

1.2 Where each dataset came from

Dataset	Source	Local path
Junyi ktbd-junyi	http://base.ustc.edu.cn/data/ktbd/junyi/	data/junyi_ktbd/
Junyi timed rows	DataShop dump → junyi.rar → timed_interactions.npz	data/ (npz stream)
ASSISTments 2009 corrected	EduData USTC skill-builder corrected	data/assistments2009/.../skill_builder_data_corrected.csv
XES3G5M	NeurIPS 2023 release (question_level CSVs / parquet)	data/xes3g5m/

1.3 What the primary dataset contains

File	Contents (features / schema)	Counts we use
vertex_id2idx	English exercise slug → integer id 0...834	835 concepts
prerequisite.json	Directed [parent, child] expert edges	978 unique edges
similarity.json	[u, v, weight] official similarity pairs	1,954 pairs
train.json	Per-line [[item_id, correct], ...] sequences	27,434 seq (len 12–200)
test.json	Same schema, published hold-out	6,290 sequences
Train interactions	Sum of filtered train.json lengths	2,342,784 attempts

File	Contents (features / schema)	Counts we use
Train+test interactions	train.json + test.json after len filter	2,842,883 attempts
Topic-level n	train+test consecutive transitions	2,809,159 pairs

Topic-level n identity (paper_junyi_counts.json): train+test interactions – train seq – test seq = 2,809,159 (equal=True). Formula: 2,842,883 – 27,434 – 6,290 = 2,809,159 when interactions=train+test.

Targets: at each cut, (1) next exercise id (ranking label); (2) Phase 3 memory = whether the **next revisit attempt** on a concept already seen in the sequence is correct (0/1). The first attempt on a concept emits **no memory row**. Attempt-level accuracy ≈ 0.51 on a 4k-sequence sample is a different task (every attempt, including firsts) and is not the Phase 3 label. Join: native: every log item id is a DAG node id.

1.4 Purpose — why these datasets?

Research problem (evaluation validity): do aggregate next-item metrics mostly reward **continue / revisit / advance** structure, and do conclusions change with concept granularity? Closest prior framing is next-basket reality-check work (repeat vs explore); we extend that to education. Junyi supplies traces + expert DAG in one id space. ASSISTments and XES3G5M test whether sticky next==last is platform/granularity-specific. Timed Junyi supports exploratory day-gap (H-forget) checks. Demo still emits review/advance rankings, but that is not a SOTA recommender claim.

1.5 Where each dataset is used in the implementation

Dataset	Train / fit	Test / validation	Role
Junyi train.json	Memory, gate, two ranker heads, popularity	—	Primary training corpus
Junyi test.json	Never fitted	Phase 6 R@K; gate AUC; exact-dup check	Published hold-out
vertex + embeddings	Query = mean of last 5	Same at eval	Semantic features + search
prerequisite.json	Advance-head unlock features	DAG figures; violation cost	Expert curriculum graph
similarity.json	Advance-head neighbour bit	Ablations	Official similarity
timed_interactions.npz	Learner-split forgetting + timed rankers	Paired Δ AUC; rank-reversal	Real day Δt
ASSISTments corrected	Popularity / sequence models (learner split)	Repetition + rank-reversal	Skill-level cross-check
XES3G5M	Same pattern, uid split	KC repetition (2 methods) + rank-reversal	KC-level cross-check

ASSISTments (prior JSON): 401,756 raw rows → 346,860 attempts; 2,920 learners (train 2,336 / test 584; overlap=0).

XES3G5M: 18,066 uids (train 14,453 / test 3,613; overlap=0). Reference KC next==last (method A, question→primary KC)=0.262; expand-all method B and question-level shares are in §4.6 / Follow-up 2 JSON — not repeated here.

1.6 Data cards, licences, and ASSISTments filter funnel

Authoritative provenance: **docs/data_cards/** (junyi_ktbd, junyi_timed, assistments2009, xes3g5m) with source URL, SHA-256, licence notes, counts, filters, timestamp applicability. Verify hashes with **scripts/fetch_datasets.py**. ASSISTments/XES licence text is marked UNVERIFIED until stored for public release.

Filter step	Learners	Rows
raw_csv	4,217	401,756
drop_na_user_order_correct	4,217	401,756
collapse_order_id_skills	4,217	346,860
keep_learners_len_ge_12	2,920	339,305
truncate_each_learner_to_200	2,920	217,645

From probe_freeze.json: 4217→2920 is len>=12 (+truncate 200), not skill collapse. sequence_builder_n_learners=2920.

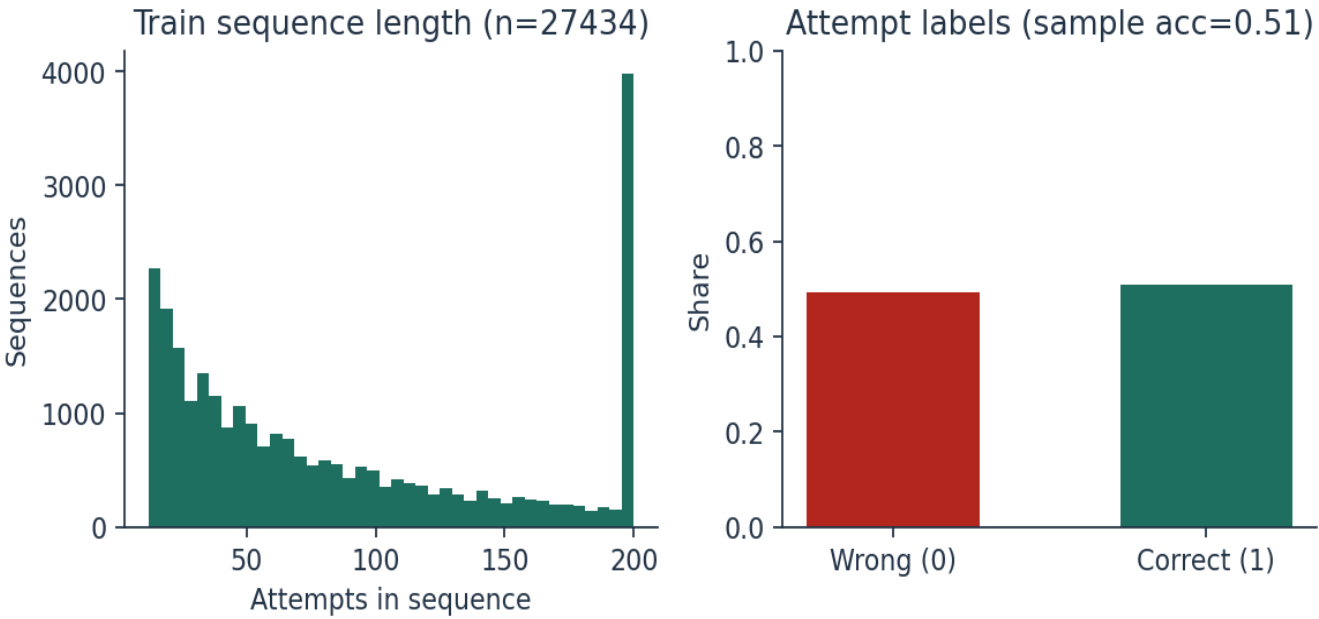


Figure 1. Left: filtered Junyi train sequence lengths. Right: attempt-level correct/incorrect balance.

2. Input at each phase (actual implementation)

Actual implementation flow (not a generic ML cartoon)

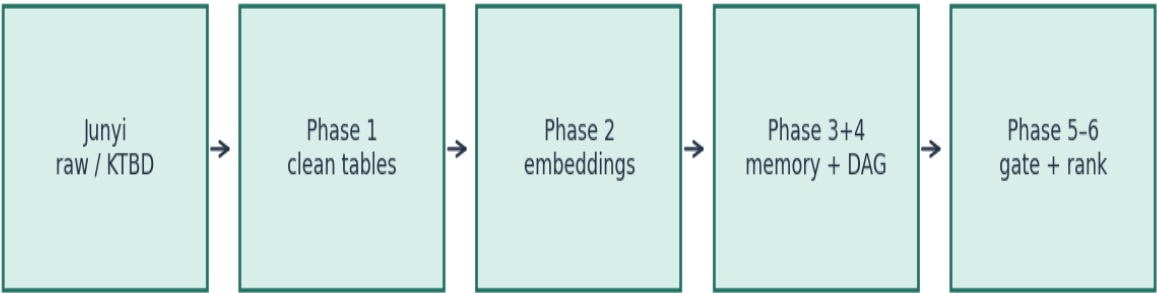


Figure 2. End-to-end flow as implemented in run_all_phases.py / serve engine.

Phase 1 — Prepare

	What
Input (where from)	Raw ktbd-junyi files in data/junyi_ktbd/ (vertex_id2idx, prerequisite.json, train/test.json).
Processing / operation	Length filter 12–200; build English concept table and edge list; confirm native id join; record faults closed vs OULAD×MOOCCubeX.
Output → next phase	concepts_junyi.csv (835), prerequisite_edges_junyi.csv (978), phase1_prepare.json. 27,434 / 6,290 sequences; 2,342,784 train interactions (train+test 2,842,883; topic n=2,809,159). → feeds Phase 2 names and Phase 3–6 sequences.

Phase 2 — Embeddings

	What
Input (where from)	835 English slugs from Phase 1 concept table.
Processing / operation	Encode with all-MiniLM-L6-v2, L2-normalise. Controls: random pairs, same-topic pairs, lexical Jaccard.
Output → next phase	concept_embeddings.npy (835×384). Linked mean cosine 0.546 vs random 0.167; pair-AUC vs random 0.904; vs same-topic 0.464; lexical AUC 0.746. → query vectors for Phase 5 heads.

Phase 3 — Memory (classification)

	What
Input (where from)	train.json (fit with sequence-id split, leakage overlap=0); official test.json for headline AUC; timed_interactions.npz for learner-split day Δt. Follow-up 3: gap≥1 day rows only.
Processing / operation	Fit mean, success_rate, HLR, logistic(+step / +day / +log1p gap / gap bins). Paired learner bootstrap on ΔAUC.
Output → next phase	test.json AUC: logistic_step 0.864 [0.861, 0.867] > success 0.858 > HLR 0.789. Follow-up 3 forgetting status= unresolved (Δ log1p-success=0.0003 [-0.001, 0.002]). → p_recall features for Phase 5.

Phase 4 — Graph / next-item mix

	What
Input (where from)	prerequisite.json + prefixes of train sequences.
Processing / operation	Build DiGraph; soft unlock = mean parent mastery; label true next as continue/revisit/advance (and hard-gate violation).
Output → next phase	DAG acyclic=True, longest path=65. Mix (n=5000): review 0.898, unlocked advance 0.020, hard-gate violation 0.082. → unlock / DAG features for Phase 5.

Phase 5 — Review–Advance ranker (RAGR)

	What
Input (where from)	Embeddings + Phase 3 memory + DAG + similarity + train prefixes.
Processing / operation	Train P(review history). Review head scores seen items; advance head scores unseen. score = P(review)·s_rev + (1-P(review))·s_adv.
Output → next phase	serve_models.joblib; train mode-review rate 0.922; review_head_n=15563, advance_head_n=1880. → frozen models for Phase 6 and the live demo.

Phase 6 — Evaluation

	What
Input (where from)	Held-out test sequences; canonical exploratory subsample 800 seq × 5 cuts; full-test replication in paper_ranking_full.json; bootstrap over sequences.
Processing / operation	Recall@5 / R@1 / MRR; force_review / force_advance; last_item on all_queries pool; pop / DAG / sim; gate AUC + PR-AUC; RAGR-no-DAG ablation; Markov order-1. Primary capped table is §4.4 from review_ranking.json; full test in §4.4b.
Output → next phase	Canonical capped RAGR R@5=0.891 [0.880, 0.902]; share(next==last)=0.803; gate AUC=0.749, PR-AUC=0.955. Full-test RAGR R@5=0.889 [0.885, 0.893] (n=6290).

Paper eval-validity pass

	What
Input (where from)	scripts/run_paper_eval.py → paper_*.json (counts, XES methods, full ranking, calibration, sensitivity, full rank-reversal).
Processing / operation	No cosmetic inflation: capped and full both reported; deviations logged in docs/DEVIATIONS.md.

	What
Output → next phase	paper_junyi_counts / paper_ranking_full / paper_calibration / paper_sensitivity / paper_rank_reversal_full / paper_xes_kc_methods. LaTeX folder: paper_assets/.

Follow-up 3 — Cross-dataset + forgetting redo + rank reversal (pre-prereg exploratory)

	What
Input (where from)	Timed Junyi (learner split); ASSISTments corrected; XES3G5M; gap≥1 day rows; 6,290-sequence exact-dup chance draw.
Processing / operation	Concept-level next==last; two independent XES KC methods; forgetting log1p/bins; popularity / recency / Markov / GRU probe / gru4rec_style / attention probe (DAG features Junyi-only) with paired sequence bootstrap.
Output → next phase	followup3_*.json + paper_rank_reversal_full.json. Forgetting status from JSON. Probe names are not published GRU4Rec/SASRec; RecBole not installed.

Probe freeze — equal-budget validation tuning (pre-prereg)

	What
Input (where from)	Learner-disjoint sequences from timed Junyi / ASSISTments / XES; data cards complete first.
Processing / operation	24-config grid for trainable probes (GRU + Attn only; RAGR-lite excluded from that claim; GRUPROBE ignores dropout). Parameter-free popularity/recency; Markov order-1 light baseline; hash test uids without scoring test (test_touched=false).
Output → next phase	probe_freeze.json. Confirmatory protocol §4.13 + docs/PREREGISTRATION*.md.

Confirmatory freeze-partition replication (after prereg-v1 + OSF)

	What
Input (where from)	Freeze train+val learners; frozen winner configs + best_epoch; cluster maps; GRU seeds 20261101/02/03.
Processing / operation	Fit on train+val (no early stop on test); score freeze test once (--touch-test + lock); two-method cluster agreement; Bonferroni m≤6; secondary common-query analysis.
Output → next phase	freeze_partition_replication.json + replication_per_seq_r5_*.json. Touch only after tag.

3. Implementation plan (dataset travel)

Junyi ktbd → length filter (Phase 1) → embed 835 names (Phase 2) → per-attempt memory rows + model select (Phase 3) → attach DAG/similarity + next-item mix (Phase 4) → train gate + two heads (Phase 5) → rank on test prefixes with sequence bootstrap (Phase 6) → serve same joblib in demo / Follow-up 3 cross-checks.

- Request input: history [(exercise, correct), ...], minimum length 2.
- Processing: rebuild attempts, success rate, Δt (steps or days), last-5 embedding mean, parent mastery.
- Output: P(review), mode □ {review, advance}, top-k items with kind, p_r recall, parents, subgraph JSON.
- Train.json fits every learned weight. Test.json is only for Phase 6 / published hold-out checks. The demo loads serve_models.joblib and never trains at click time.
- ASSISTments / XES3G5M never enter the shipped Junyi joblib; they only validate generality of repetition and ranking-order claims.

4. Output / results obtained

4.1 Files the implementation writes

Path	What it is
outputs_junyi/concepts_junyi.csv	835 English concepts + degrees

Path	What it is
outputs_junyi/prerequisite_edges_junyi.csv	978 learn-source-before-target rows
outputs_junyi/concept_embeddings.npy	(835, 384) float32, L2 unit
outputs_junyi/phases/phase1–6_*.json	Phase contracts (machine-readable)
outputs_junyi/phases/followup3_*.json	Forgetting, concept table, dups, rank-reversal
outputs_junyi/phases/paper_*.json	Full-test ranking, calibration, sensitivity, XES methods
paper_assets/	LaTeX tables (.tex) + vector figures (.pdf)
outputs_junyi/serve_models.joblib	Logistic memory + gate + two heads
outputs_junyi/figures/*.png	Distributions, ROC, mix, ranking, DAG, flow
docs/NeuroTrace-DAG_Data_Flow_Report.pdf	This document (built from JSON)

4.2 Sample live prediction (product output)

History: one step equations (correct) → one step inequalities (correct) → solving quadratics by factoring (wrong).
Categories: **review** (seen) vs **advance** (unseen).

Rank	Exercise	Kind	Score	p [*] recall	Parents
1	solving quadratics by factoring	review	2.81	0.08	factoring polynomials 1
2	one step inequalities	review	0.22	0.92	inequalities on a number line
3	one step equations	review	−0.52	0.93	evaluating expressions 1
4	addition 1	advance	−7.25	0.55 unseen	root
5	number line	advance	−7.34	0.55 unseen	root

P(review)=0.86, mode=review. Last item failed → restudy ranks first (low p^{*} recall 0.08). Demo head-score calibration (softmax temperatures) is deferred; do not treat raw decision_function mixing as a calibrated probability model.

4.3 Phase 4 next-item mix (intermediate output)

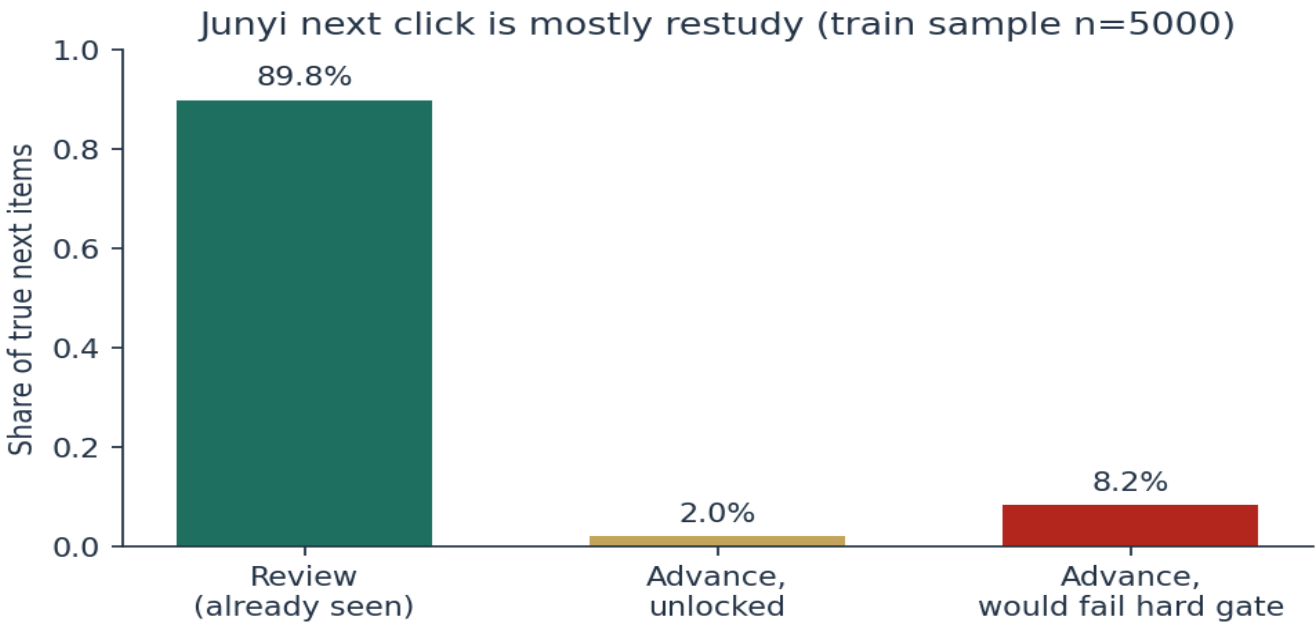


Figure 3. Train-prefix mix (n=5000): review 0.898, unlocked advance 0.020, hard-gate violation 0.082. Monolithic next-item scores on this log mostly measure restudy.

Source (missing prereq)	Target attempted	Count
count number to 100	addition 2	7

Source (missing prereq)	Target attempted	Count
understand coins	addition 2	7
number line 4	addition 2	7
adding and subtracting within 20	addition 2	7
division 2	division 2.9	6
divisibility intuition	divisibility tests	6

Table: most frequent hard-gate skips in the Phase 4 sample.

4.4 Primary ranking table (from review_ranking.json only)

Eval: 800 seq × 5 cuts (canonical **capped** exploratory subsample; full test in §4.4b). seed=0. Action mix continue/bounce/far/advance = 0.803/0.044/0.043/0.110. **last_item** below = score-1-on-last on **all_queries** pool (not a review-only pool).

Canonical ranking (review_ranking) · RAGR R@5=0.891 [0.880, 0.902] · 4000 queries

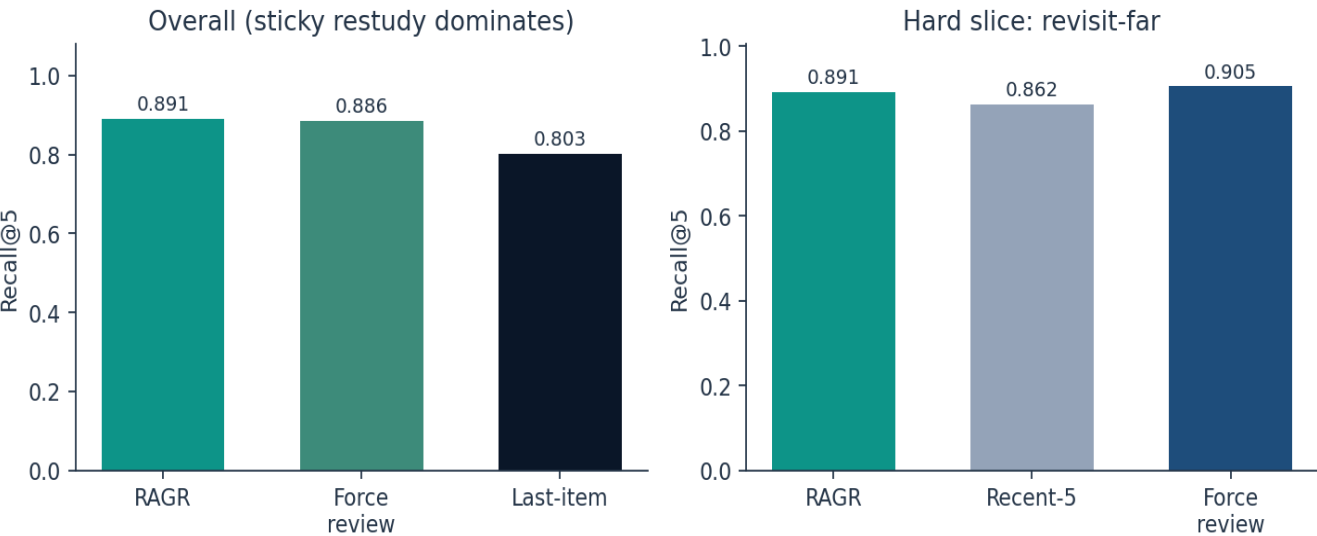


Figure 4. Ranking figure assets; numeric table is exclusively from review_ranking.json.

Method	R@1 (seq. boot. CI)	R@5 (seq. boot. CI)	MRR (seq. boot. CI)	Pool
RAGR	0.802 [0.786, 0.819]	0.891 [0.880, 0.902]	0.845 [0.831, 0.858]	all_queries
Force review	0.803 [0.787, 0.818]	0.886 [0.875, 0.897]	0.841 [0.828, 0.854]	all_queries
Recent-5	0.803 [0.785, 0.818]	0.885 [0.872, 0.897]	0.840 [0.826, 0.853]	all_queries
last_item	0.803 [0.786, 0.819]	0.803 [0.787, 0.819]	0.804 [0.787, 0.819]	all_queries
Frequency	0.468 [0.445, 0.488]	0.827 [0.812, 0.840]	0.621 [0.603, 0.637]	all_queries

CI's are sequence-level bootstrap (mean of per-sequence means, n_boot=1000) from review_ranking.json. The older Phase 6 print 0.891 [0.879, 0.905] (phase6_eval.json ragr_all, CI [0.8794875, 0.90475]) is the same point estimate under a different bootstrap draw and is **superseded** by the review_ranking.json interval [0.880, 0.902].

Paired Δ R@5	Mean	95% CI
RAGR – force_review	0.0052	[0.003, 0.008]
RAGR – recent5	0.0067	[0.004, 0.009]

Note: RAGR – last_item is omitted here (last_item on all_queries is a sticky continue ceiling, not a matched review-pool baseline). See review_ranking.json if needed for audit.

4.4b Full-test ranking (paper_ranking_full.json)

Exploratory summary (pre-prereg, CI verdict rule t_R@5=0.01 / t_AUC=0.005): no slice reversals flagged under the required-model H-rev rule in exploratory data; only XES native vs cluster flips (GRU–Markov sign change with both sides supported); Markov beats RAGR on advance (supported_B); DAG features negligible vs RAGR; Phase 3 logistic vs success inconclusive at t=0.005. See

paper_paired_verdicts.json → exploratory_summary_for_pdf.

Full published test.json: 6290 sequences × 5 cuts = 31,450 queries; seed=0; sequence bootstrap n_boot=1000. Not tuned to look better than the capped table — report both.

Method	R@5 (seq. boot. CI)	n seq
RAGR	0.889 [0.885, 0.893]	6290
RAGR-no-DAG	0.889 [0.884, 0.893]	6290
Markov order-1	0.856 [0.850, 0.861]	6290
Recent-5	0.880 [0.876, 0.885]	6290
RAGR (advance)	0.081 [0.071, 0.091]	2554
RAGR-no-DAG (advance)	0.080 [0.069, 0.089]	2554
Markov (advance)	0.242 [0.226, 0.257]	2554

RAGR-no-DAG zeros advance unlock + is_child features (cols 1, 3). Advance-slice R@5 is the validity-relevant contrast: sticky overall R@5 is not a recommender win.

4.4c Calibration (gate + $\hat{p}_{\text{recall}}$)

Predictor	n	Base rate	Brier	ECE	Δ Brier vs base
Gate P(review)	31,450	0.886	0.0928	0.0335	-0.0085
$\hat{p}_{\text{recall}}$ (continue queries)	24,809	0.474	0.1480	0.0410	-0.1013

Δ Brier vs base = Brier(model) – Brier(constant base-rate); negative $\square$ better than base rate. For $\hat{p}_{\text{recall}}$, base rate=0.474 so Brier(base)=base×(1–base)=0.2493; Δ Brier=-0.1013. Lowest predicted bin [0.000, 0.100): n=2219, pred=0.0909, obs=0.0721, obs–pred=-0.0188. Reliability plots: paper_assets/figures/fig_gate_reliability.pdf, fig_memory_reliability.pdf.

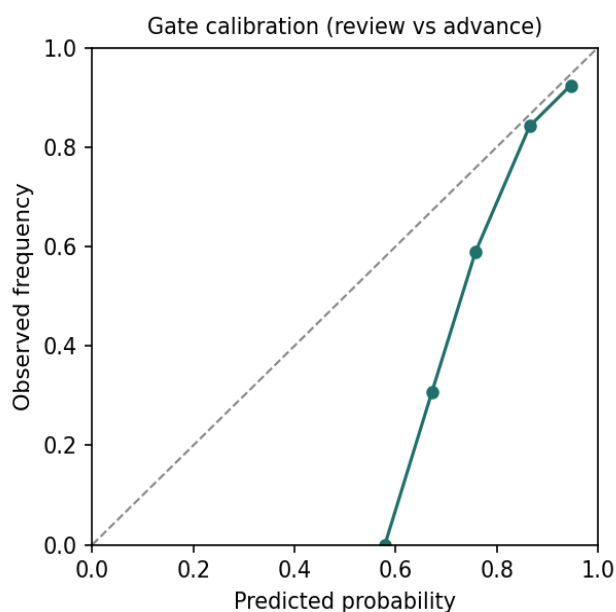
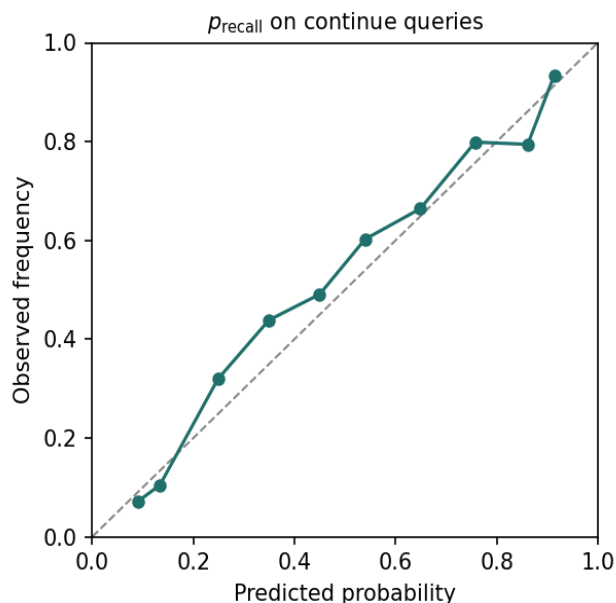


Figure 4b. Gate reliability diagram.

Figure 4c. p^*_recall reliability on continue queries.

4.4d Sensitivity (length cap, cuts, timed-6k selection)

Max len	n train		n test		Topic share(next==last)
100	27434		6290		0.9245
200	27434		6290		0.9300
400	27434		6290		0.9300
Cuts / seq	continue	revisit	advance	n queries	
3	0.791	0.084	0.125	2400	
5	0.801	0.088	0.111	3938	
10	0.794	0.100	0.105	7479	

Timed 6k selection bias: first-6k file-order share=0.6344 (n=405309); random-6k=0.6350 (n=409120); $\Delta(\text{first}-\text{random})=-0.0006$ (eligible users=72457).

4.5 Concept-level repetition (Follow-up 3) — exercise/question rows dropped

Dataset / unit	next==last	n	continue	revisit	advance
Junyi topic	0.930	2,809,159	0.930	0.040	0.030
ASSISTments composite skill token	0.679	214,725	0.679	0.184	0.136
XES3G5M KC (method A: question→primary KC)	0.262	3,975,344	0.262	0.313	0.425
XES3G5M KC (method B: expand-all KC tags)	0.216	4,428,696	0.216	0.365	0.419
XES3G5M KC (method B parquet is_repeat=0)	0.263	4,410,668	0.263	0.302	0.435
ASSISTments primary atomic skill	0.696	214,725	—	—	—

Note from JSON: exercise/question-level rows removed from cross-dataset comparison XES method B in this table is expand-all KC tags (share≈0.216), not the deprecated first-KC-only B that duplicated A. Native-unit cross-dataset shares are descriptive until cluster-matched H-rep. ASSISTments truncation to 200 dropped 121,660 of 339,305 rows after len≥12.

4.6 XES3G5M KC repetition — both methods recorded

Method	Rule	next==last	n	continue / revisit / advance
A (primary reported)	map each question to its first KC, then share(next==last) on KC ids	0.262	3,975,344	0.262 / 0.313 / 0.425
B (expand-all KC tags)	expand every KC tag on each question into the stream (F2 expand-all). Independent of method A — not 'first KC only'.	0.216	4,428,696	0.216 / 0.365 / 0.419

Methods A and B are **independent** (numerically_identical=False). primary_reported=method_a for concept-level table; method_b expand-all as sensitivity. Old first-KC-only method B is deprecated.

4.7 Forgetting redo (gap ≥ 1 day only) — status unresolved

Protocol: gap >= 1 calendar day only; split=learner-level 80/20; feature=log1p(gap_days). Learners train/test=35,997/8,999; rows=509,863/125,637.

Model		AUC	
success_rate only (logistic)		0.661	
+ log1p(gap)		0.661	
+ gap bins		0.662	
Paired ΔAUC (learner bootstrap)	Mean	95% CI	CI excludes 0?
log1p – success	0.0003	[-0.001, 0.002]	False
gap bins – success	0.0010	[-0.000, 0.002]	False

Status=**unresolved** (rule: log_gap_helps if paired CI of (AUC_log - AUC_success) entirely above 0; log_gap_hurts_vs_success if entirely below 0; else unresolved). Do not call this a null result.

Accuracy by gap bin within success-rate strata (test rows, gap≥1):

Success stratum	Gap bin	n	Accuracy	Mean success rate
0	1_3_days	8,566	0.639	0.308
0	4_14_days	9,166	0.637	0.308
0	15p_days	7,527	0.660	0.292
1	1_3_days	6,293	0.804	0.739
1	4_14_days	10,052	0.819	0.747
1	15p_days	9,310	0.789	0.751
2	1_3_days	14,171	0.886	0.982
2	4_14_days	28,710	0.886	0.982
2	15p_days	31,842	0.879	0.983

4.8 Exact-duplicate chance baseline

Check	n	Exact-hash overlaps
Real train ∩ test	6,290 test seq	23
Chance: pseudo-test from train	6,290 vs 21,081 remaining	23
Real – chance	—	0

Real overlap (23) equals chance (23) under seed=0. The 23 exact duplicates are consistent with chance, not evidence of a special train/test leak.

4.9 Rank-reversal study (paper_rank_reversal_full.json)

EXPLORATORY — not confirmatory. prereg-v1 existed before run_paper_eval? **False**. n_test sizes match freeze 15% ({'junyi_timed': True, 'assistsments': True, 'xes3g5m': True}); uid hashes match freeze? **False** (paper seed=0, freeze seed=20261004). Confirmatory replication reuses the **freeze holdout** (seed 20261004), not a new learner-split seed — see §4.13.

Methods on the same learner-disjoint split (seed=0, cuts=5, full_test=True): popularity, recency, **Markov order-1**, GRU probe, gru4rec_style robustness check, attention probe. **Recency on advance = N/A** for advance-slice display (counts as 0 in all/nonrepeat aggregates under the registered replication rule). Primary confirmatory H-rev is native↔cluster GRU–Markov (§4.13), not slice order alone. RecBole not installed. Freeze val numbers in §4.11.

junyi_timed — R@5 on all queries

Rank	Method	R@5 (all)	R@5 (advance)
1	Markov order-1	0.800	0.417
2	GRU4Rec-style local probe (not RecBole)	0.797	0.407
3	GRU-based next-item probe	0.790	0.397
4	attention-based next-item probe	0.707	0.259
5	recency	0.706	N/A
6	popularity	0.191	0.157
Slice	Kendall τ vs all	Slice order (excl. recency on advance)	
revisit	0.200	recency > GRU4Rec-style local probe (not RecBole) > GRU-based next-item probe > Markov order-1 > attention-based next-item probe > popularity	
advance	1.000	Markov order-1 > GRU4Rec-style local probe (not RecBole) > GRU-based next-item probe > attention-based next-item probe > popularity	
nonrepeat	0.867	GRU4Rec-style local probe (not RecBole) > Markov order-1 > GRU-based next-item probe > attention-based next-item probe > recency > popularity	

H-rev advance (required models excl. recency): ordering_changed=False; Kendall τ =1.000. Per-sequence bootstrap flip probabilities require stored per-seq R@5; not available from the original paper_rank_reversal_full aggregates. Confirmatory replication will store per-seq scores and report P(order flip).

Split: unit=exercise, train/test=5100/900, eval_seq=900; status=EXPLORATORY.

Local GRUProbe emb=128 hidden=128 epochs=5 — published-style robustness check; RecBole not installed in this environment (see DEVIATIONS).

assistments — R@5 on all queries

Rank	Method	R@5 (all)	R@5 (advance)
1	GRU4Rec-style local probe (not RecBole)	0.933	0.654
2	GRU-based next-item probe	0.918	0.552
3	Markov order-1	0.887	0.510
4	attention-based next-item probe	0.883	0.430
5	recency	0.857	N/A
6	popularity	0.428	0.214
Slice	Kendall τ vs all	Slice order (excl. recency on advance)	
revisit	0.333	recency > GRU4Rec-style local probe (not RecBole) > GRU-based next-item probe > attention-based next-item probe > Markov order-1 > popularity	
advance	1.000	GRU4Rec-style local probe (not RecBole) > GRU-based next-item probe > Markov order-1 > attention-based next-item probe > popularity	
nonrepeat	0.867	GRU4Rec-style local probe (not RecBole) > GRU-based next-item probe > attention-based next-item probe > Markov order-1 > recency > popularity	

H-rev advance (required models excl. recency): ordering_changed=False; Kendall τ =1.000. Per-sequence bootstrap flip probabilities require stored per-seq R@5; not available from the original paper_rank_reversal_full aggregates. Confirmatory replication will store per-seq scores and report P(order flip).

Split: unit=composite_skill_token, train/test=2481/439, eval_seq=439; status=EXPLORATORY.

Local GRUProbe emb=128 hidden=128 epochs=5 — published-style robustness check; RecBole not installed in this environment (see DEVIATIONS).

xes3g5m — R@5 on all queries

Rank	Method	R@5 (all)	R@5 (advance)
1	Markov order-1	0.733	0.727
2	GRU4Rec-style local probe (not RecBole)	0.693	0.689
3	GRU-based next-item probe	0.619	0.612
4	attention-based next-item probe	0.598	0.589
5	recency	0.030	N/A
6	popularity	0.015	0.011

Slice	Kendall τ vs all	Slice order (excl. recency on advance)
revisit	0.867	Markov order-1 > GRU-based next-item probe > GRU4Rec-style local probe (not RecBole) > attention-based next-item probe > recency > popularity
advance	1.000	Markov order-1 > GRU4Rec-style local probe (not RecBole) > GRU-based next-item probe > attention-based next-item probe > popularity
nonrepeat	1.000	Markov order-1 > GRU4Rec-style local probe (not RecBole) > GRU-based next-item probe > attention-based next-item probe > recency > popularity

H-rev advance (required models excl. recency): ordering_changed=False; Kendall τ =1.000. Per-sequence bootstrap flip probabilities require stored per-seq R@5; not available from the original paper_rank_reversal_full aggregates. Confirmatory replication will store per-seq scores and report P(order flip).

Split: unit=question, train/test=15355/2711, eval_seq=2711; role=descriptive_curriculum_order; status=EXPLORATORY.

Local GRUProbe emb=128 hidden=128 epochs=5 — published-style robustness check; RecBole not installed in this environment (see DEVIATIONS).

4.10 Readiness diagnostic (review_readiness.json)

Status=**detectable_but_negligible** (exploratory). Practical threshold ΔAUC =0.005. Primary ΔAUC mean=0.0005 95% CI=[0.000, 0.001] (ci_above_zero=True). Rule from JSON: supported if mean ΔAUC $\geq$ 0.005 and CI entirely above 0; detectable_but_negligible if CI above 0 but mean < threshold; else unresolved

4.11 Probe freeze on validation (probe_freeze.json, schema v4)

Equal-budget **trainable** grid: GRU + Attn only (RAGR-lite is **not** in that claim). **GRUProbe has no dropout** — dropout in the grid is a no-op for GRU (Attn uses it). Parameter-free: popularity, recency. Parameter-light: **markov_order1** (not on the 24-config grid). Grid configs=**24**. Selection=**val_nonrepeat_r5_then_lowest_ce_within_0.005** (tol=0.005). Seed=20261004. Test hashed only (**test_touched=false**). Attn prereg=**excluded**. H-rev primary=matched_concept_clusters (~120); native secondary; XES question-level = curriculum-order / descriptive for non-H-rev. Min models for tables: popularity, recency, markov_order1, gru_probe.

Hyperparameter	Values
lr	[0.0005, 0.001, 0.002]
emb	[32, 64, 128, 256]
dropout	[0.1, 0.3] (no-op for GRUProbe)
max_epochs / patience	10 / 2
GRU hidden / windows	max(64, emb*2); MAX_WINDOWS=20,000; ctx=50; batch=64
selection	val non-rep R@5; within 0.005 $\rightarrow$ lowest val CE

Grid note: Cartesian lr $\times$ emb $\times$ dropout = 24 configs; identical for every trainable model (GRUProbe, AttnProbe) per dataset. RAGR-lite is not part of this equal-budget trainable grid. GRUProbe has no dropout module (dropout grid is a no-op for GRU). max_epochs=10, patience=2 (see docs/DEVIATION Deviations: docs/DEVIATIONS.md).

Dataset	Native unit	Native role	Pop	Markov nat.	Markov cl.	GRU nat.	GRU cl.	n_cl used
junyi_timed	exercise	—	0.162	0.430	0.471	0.445	0.515	115

Dataset	Native unit	Native role	Pop	Markov nat.	Markov cl.	GRU nat.	GRU cl.	n_cl used
assistments	composite_skill_token	—	0.271	0.568	0.594	0.750	0.771	106
xes3g5m	question	curriculum/de sc	0.011	0.739	0.636	0.614	0.761	120

Markov order-1 is frozen at both native and cluster units (parameter-light; not on the 24-config grid). n_cl used = unique cluster ids after mapping train+val and dropping empties (115 / 106 / 120 vs target 120).

Dataset	Pop nq	Markov nat. nq	GRU nat. nq	Markov cl. nq	GRU cl. nq
junyi_timed	762	777	784	636	630
assistments	628	628	628	599	599
xes3g5m	1,989	1,985	1,987	651	679

Validation non-repeat query counts. Slice < 500 queries → N/A at replication. XES native/cluster R@5 may use a 400-sequence cap (see DEVIATIONS.md).

Dataset	Unit	lr	emb	best_ep	dropout	val nonrep R@5	val CE
junyi_timed	native	0.001	256	2	0.1	0.445	2.2640
junyi_timed	cluster	0.0005	256	3	0.1	0.515	1.4775
assistments	native	0.0005	256	6	0.3	0.750	1.1291
assistments	cluster	0.001	256	4	0.3	0.771	1.0661
xes3g5m	native	0.0005	256	4	0.1	0.614	3.8651
xes3g5m	cluster	0.0005	256	7	0.1	0.761	1.2241

Dataset	Clustering	k	Fallback?	ASSIST names?
junyi_timed	text_minilm_kmeans	120	no	n/a
assistments	text_skill_name_minilm_kmeans	120	no	yes
xes3g5m	train_only_cooccurrence_random_projection_kmeans	120	yes	n/a

XES curriculum-order (descriptive native): mode successor=0.350; global-order successor=0.196.

ASSISTments skill construction: atomic=123; composite=150.

Junyi timed 6,000-learner selection: not_random_sample=True.

Elapsed=12480.7s. min_slice_queries=500. Canonical upgrade: scripts/run_preprereg_freeze_v4.py.

4.11b Phase 3 confusion / base-rate audit

Headline 0.877 used the first 80,000 test.json memory (revisit) rows in file order and is **superseded**. Replacement: ALL 449,453 test.json memory rows (correct_rate=0.446); logistic_step_dt AUC=0.864 [0.861, 0.867] (sequence bootstrap); PR-AUC=0.826. CM @0.5 integers TN=209,734 FP=39,385 FN=56,994 TP=143,340. Task = correctness on revisit attempts only.

4.11c GRU vs Markov paired sequence CIs (validation)

H-rev asks whether a probe's ranking vs baselines reverses when the action unit changes (native vs matched clusters). Markov order-1 is the parameter-light 1-step successor table. If GRU loses to Markov at a unit, that unit's GRU 'win' vs popularity/recency is not a learned-structure win: a count table already captures the regularity (sticky or curriculum-order). A confirmatory H-rev claim at that unit cannot treat GRU as beating the required baseline set. Native XES (7,439 questions, mode successor ≈0.35, GRU 0.614 vs Markov 0.739 at freeze) is therefore descriptive. The informative contrast is the cluster unit, where GRU can still beat Markov.

Dataset	Unit	GRU R@5	Markov R@5	Δ GRU-Mk	95% CI	Winner
junyi_timed	native	0.415	0.408	0.0068	[-0.023, 0.039]	tie/NS
junyi_timed	cluster	0.514	0.480	0.0340	[0.002, 0.066]	yes
assistments	native	0.725	0.568	0.1567	[0.117, 0.196]	yes

Dataset	Unit	GRU R@5	Markov R@5	Δ GRU-Mk	95% CI	Winner
assistments	cluster	0.764	0.594	0.1699	[0.126, 0.217]	yes
xes3g5m	native	0.570	0.718	-0.1480	[-0.170, -0.128]	Markov
xes3g5m	cluster	0.765	0.618	0.1473	[0.114, 0.181]	yes

XES native longer-epoch diagnostic: max_epochs=30, patience=5, best_epoch=4, val CE=3.8841. If GRU still loses to Markov after 30 epochs / patience 5, undertraining at max_epochs=10 is not a sufficient explanation; the 7,439-way catalog is curriculum-order-structured and a 1-step table is the right null.

4.11d Dual clustering + feasible matched counts

Actual used cluster counts (freeze primary, dense remap): 115 / 106 / 120. ~40: all three datasets. ~120: all three, but ASSISTments is near-native (150 items / 106 used). ~800: Junyi and XES only; Junyi k=800 is close to its 835-item catalog; ASSISTments = N/A. A method contrast counts only if both methods agree. XES text: XES3G5M question_level CSVs expose question ids / KC tags only. No question stem text for MiniLM. Text clustering is N/A..

Target k	Junyi (835)	ASSISTments (150)	XES (7,439)
40	ok	ok	ok
120	ok	ok_near_native	ok
800	ok_near_catalog	N/A	ok

Dataset	Text?	Text method	2nd method	ARI
junyi_timed	yes	text_minilm_kmeans	train_only_cooccurrence_random_projection_kmeans	0.038
assistments	yes	text_skill_name_minilm_kmeans	train_only_cooccurrence_random_projection_kmeans	0.073
xes3g5m	no	N/A	train_only_cooccurrence_random_projection_kmeans	N/A

4.12 Inertia / pedagogy intermediate outputs

After wrong attempt, P(next=last)=0.866; after correct, 0.785. Hard-gate violations cost accuracy: unlocked 0.769 vs violate 0.668 (gap 0.101).

4.13 Confirmatory H-rev protocol (preregistration draft)

Drafts (not yet the git tag **prereg-v1**): docs/PREREGISTRATION.md, docs/PREREGISTRATION_REPLICATION.md, docs/DEVIATIONS.md (attached to OSF/Zenodo and covered by the tag). Human tags → registers → runs **scripts/run_freeze_partition_replication.py --touch-test** once (touch_test.lock.json blocks a second run unless DEVIATIONS records TOUCH_TEST_OVERRIDE:).

Primary claim: dataset D satisfies H-rev iff both unit verdicts are supported_* with **opposite signs**; project claim = at least one of {Junyi, ASSISTments, XES}. Family **m_planned = 6** (3 datasets × native/cluster); Bonferroni level 1–0.05/m. Two-method Junyi/ASSIST cluster cells use an **intersection rule** (one family member, not two). Cell N/A if <500 non-repeat queries → m := 6 – n_NA. Pipeline written to JSON: eligibility → m → CIs → dataset claims → project claim.

Item	Registered value
Holdout	freeze seed 20261004 (70/15/15); not junyi_pipeline.SEED=0
GRU seeds	20261101, 20261102, 20261103; per-seq R@5 = mean; bootstrap on mean
Markov	deterministic single fit
Bootstrap	≥10,000 sequence-level; zlib.crc32 seeds (not builtin hash)
Clustering	used counts 115 / 106 / 120; seed 20261021; ARI 0.038 / 0.073
Two-method	text □ cooc; cooc uses primary cluster GRU config (no retune); XES cooc only
Recency	0 on advance for all/nonrepeat; N/A only for advance-slice display

Item	Registered value
recall_at ties	lower item index wins among equal scores (stable lexsort)
Secondary	common native□cluster non-repeat queries; same CI rule; primary-only flips labeled
H-rep / graph / forget	no confirmatory H-rep; graph & forgetting exploratory Junyi-only
First-order Markov limit	stronger sequence baselines could change conclusions

Pre-tag dual-map GRU–Markov on freeze **val** (not freeze test):

Dataset	text verdict	cooc verdict	registered cluster	ARI
junyi_timed	inconclusive	supported_A	inconclusive	0.038
assistments	supported_A	supported_A	supported_A	0.073

Exploratory paired-verdict blurb: {'slice_reversals_flagged_under_h_rev_required_models': False, 'xes_native_vs_cluster_sign_flip': True, 'xes_primary_h_rev_supported': True, 'markov_beats_ragr_on_advance': True, 'dag_features_negligible': True, 'phase3_logistic_vs_success': 'inconclusive'}

Seed-0 exploratory test ∩ freeze test overlap (still EXPLORATORY): Junyi 128/900 (14.2%); ASSISTments 70/439 (15.9%); XES 401/2711 (14.8%).

freeze_partition_replication.json: schema=freeze_partition_replication_v3; touch_test=False; has_unit_scores=False. Confirmatory test numbers appear only after --touch-test post-tag.

Locked SHA-256 digests (prereg_v1_locked_hashes.json; pytest fails if these files change):

Locked file	SHA-256
probe_freeze.json	28a8248a1664f0e3dcf2a833be3e20f8e2b94bd457d1f4073f3e45095e4ffaff
run_freeze_partition_replication.py	09b419a5aaac1326965ef03826e04eae3c05ecb4cb45070ab50167f6c394fe
dual_clustering.json	6c98f9e301ea0566165b492bdc179d9b2444f62504cc2de548778b48a2517923
junyi text/item maps	be9981bf6d7679bce273a4689937ef20d60d1257fc0897d1d7fd7179daf6c316
junyi cooc map	5dfc3e134cd5d64db4c7a5224a026be162a5db78054455c4fe617ccffa44b69e
assist text/item maps	7676e8dbe9a90c309c3e2578526c81f85688ef1088dd6a8c7327b74596225327
assist cooc map	e82d2ba2217a792b4fe4ed3ce45b5c2fe5634d258c27934c9f02dccc69e6a159
xes cooc/item maps	097696754951b49fceaa4369ee45f4cd7adf6dae75ec6c8cc93ade21fa5f018d

5. Visualizations

Figures below are generated artifacts under outputs_junyi/figures/ and embedded here. They show data distributions, embedding structure, DAG neighbourhoods, next-item mix, ranking, memory AUC, ROC, and gate regret — not decorative placeholders.

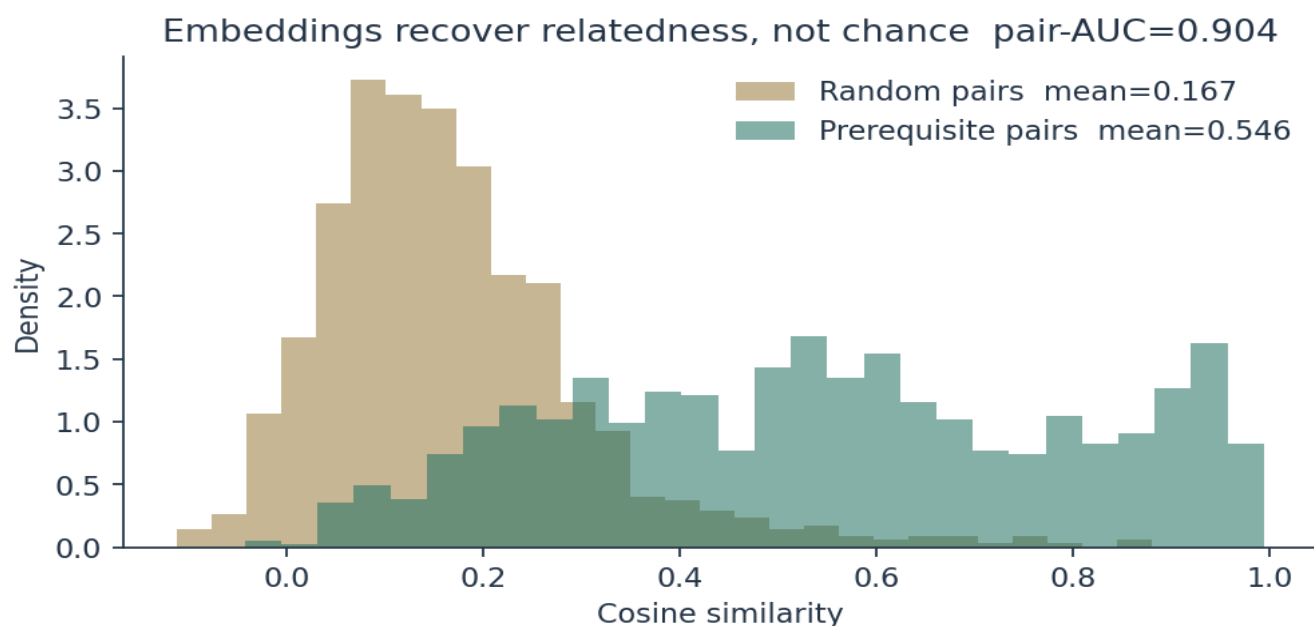


Figure 5. Embedding cosine: linked 0.546 vs random 0.167 (pair-AUC 0.904). Same-topic control pair-AUC 0.464.

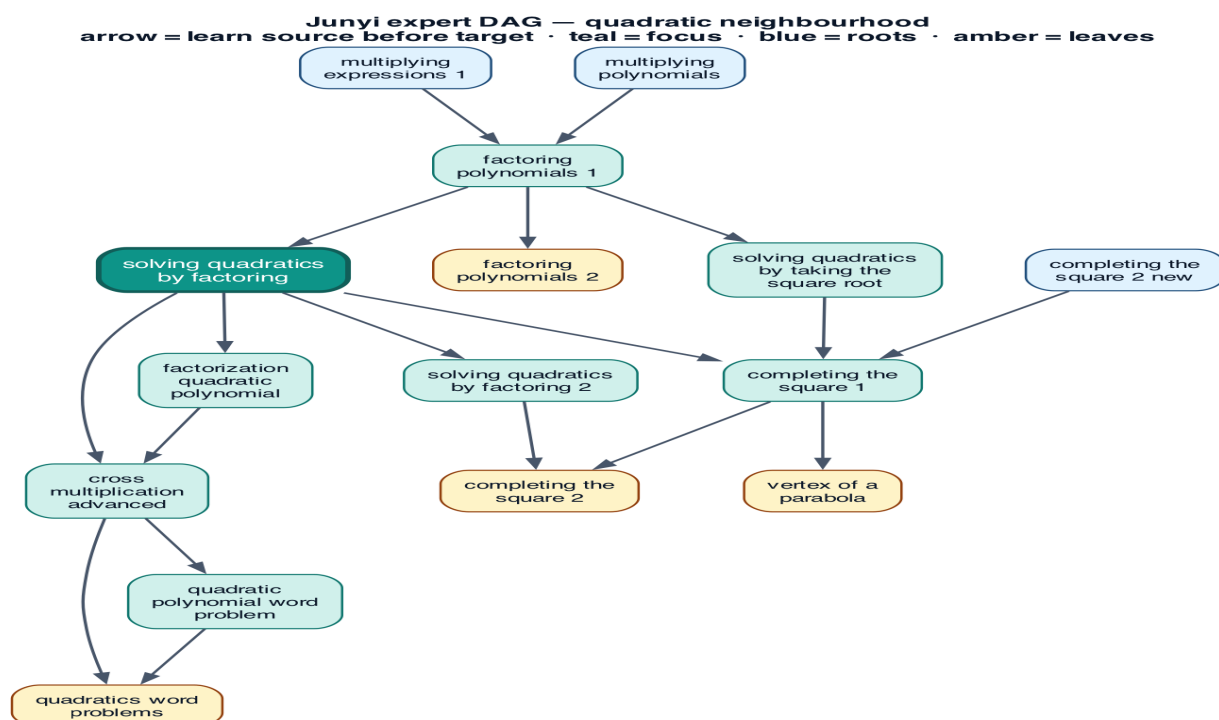


Figure 6. Expert DAG — quadratic neighbourhood. Arrow = learn source before target.

Junyi expert DAG · 978 edges · acyclic=True

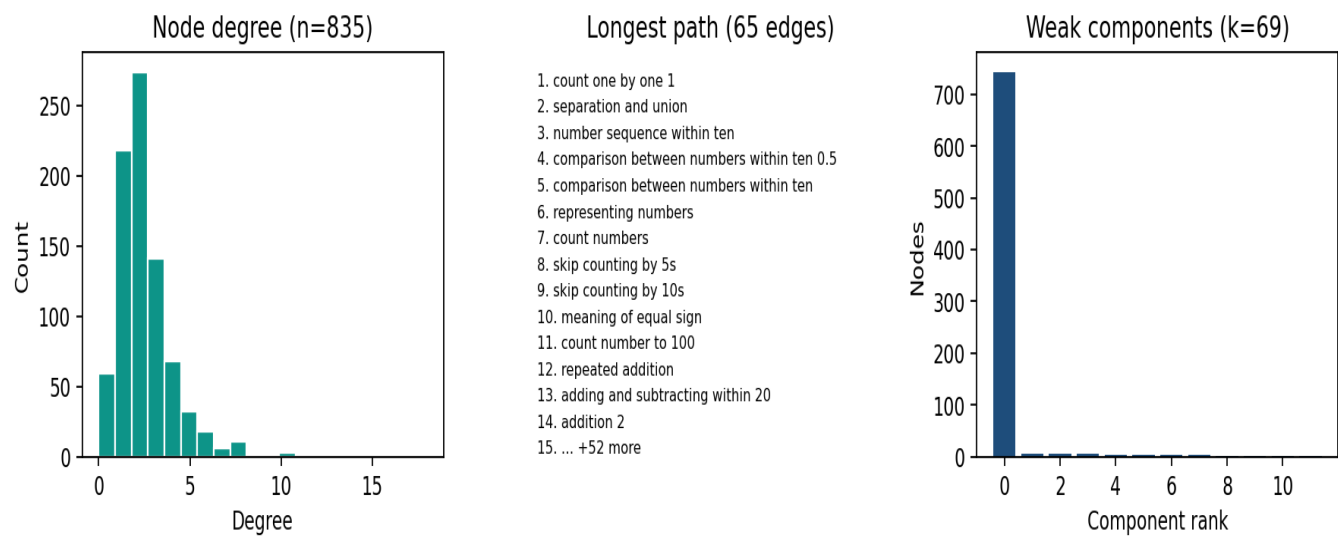


Figure 6b. Full DAG structure: degrees, longest-path sample, component sizes.

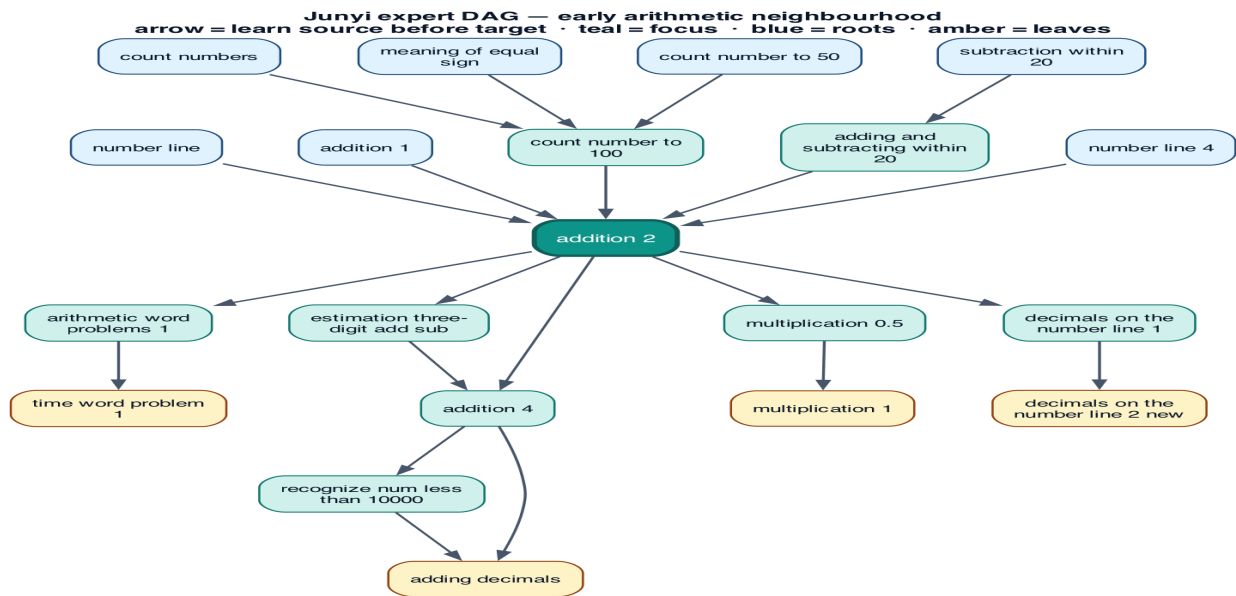


Figure 6c. Early arithmetic neighbourhood (same layout rules).

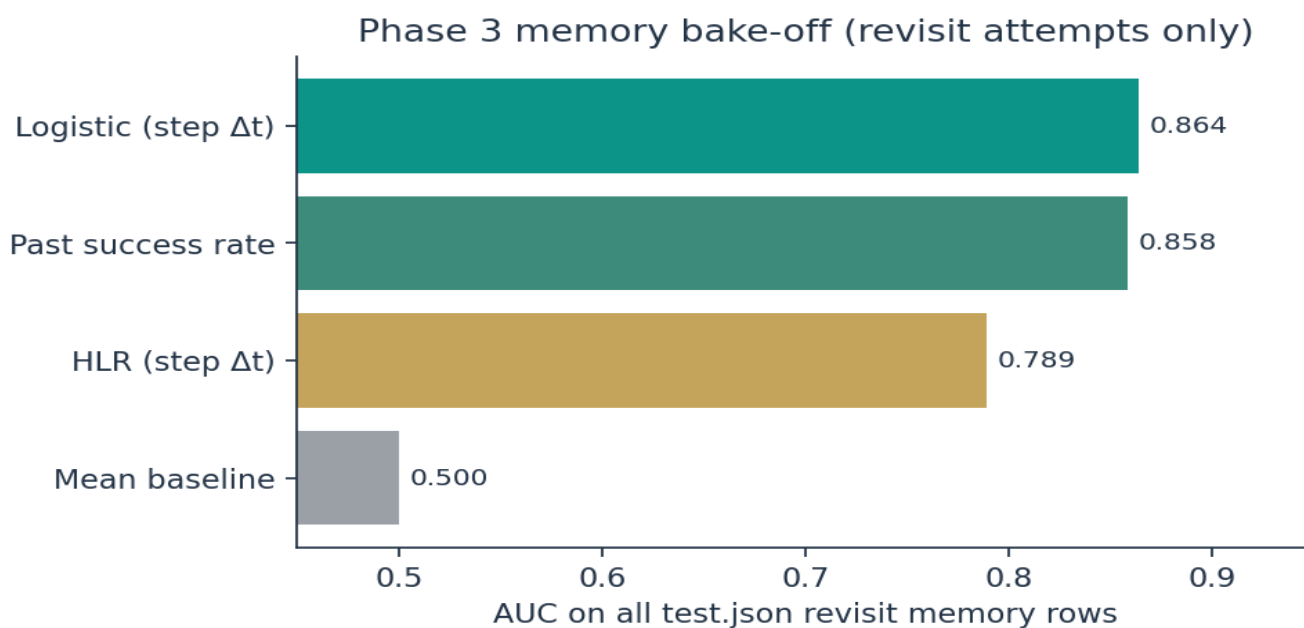


Figure 7. Memory AUCs on official test.json all 449,453 test.json memory (revisit) rows: logistic 0.864 [0.861, 0.867] / success 0.858 / HLR 0.789. Task = correctness on revisit attempts only. CIs are sequence-level bootstrap once phase3_memory_fulltest.json is the source.

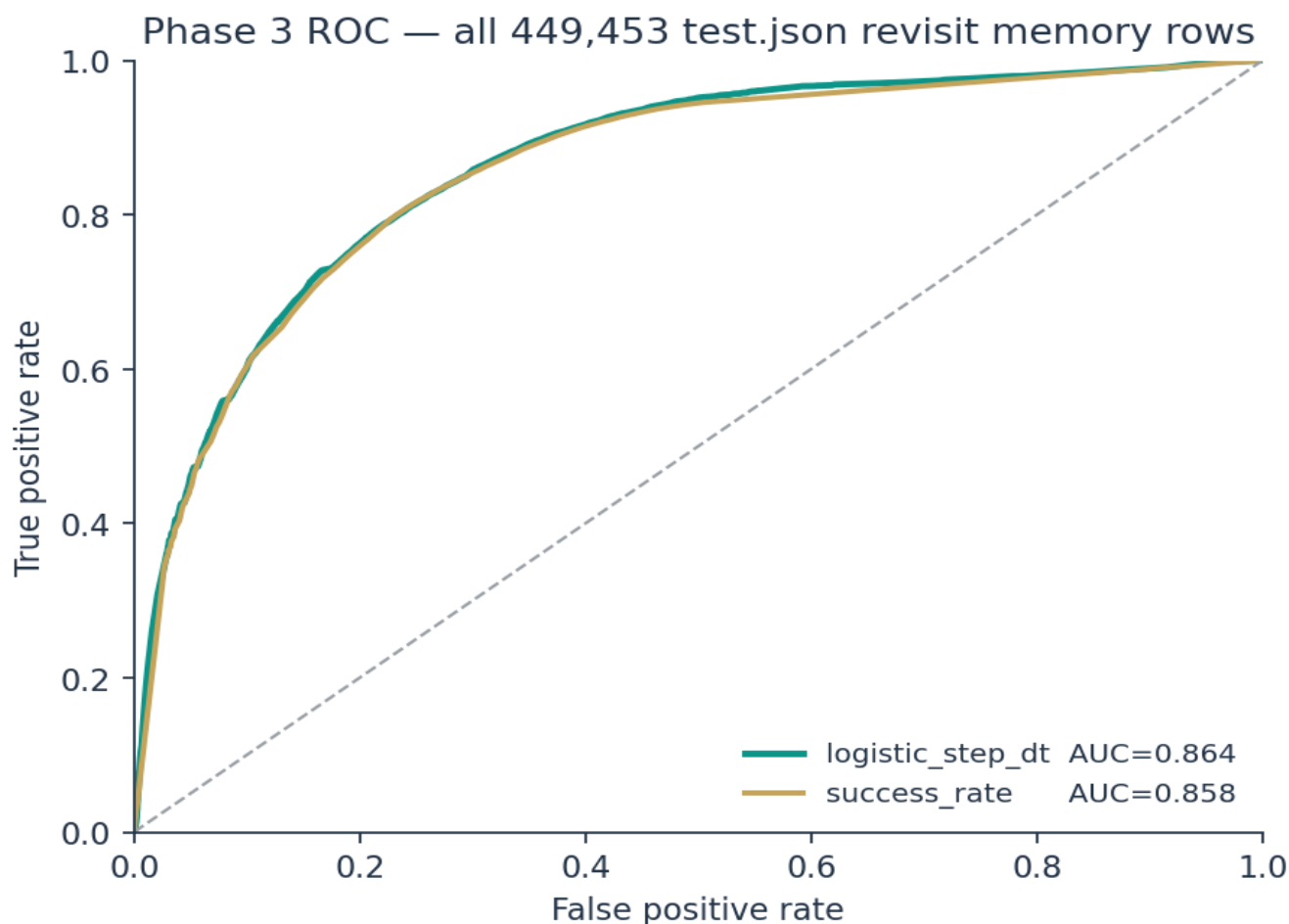


Figure 7b. ROC on all 449,453 test.json memory (revisit) rows — logistic_step_dt vs success_rate (predictions = $P(\text{correct on this revisit})$; labels = revisit-attempt correctness). The first attempt on a concept is not a row.

Phase 3 logistic @0.5 — all 449,453 test.json revisit memory rows

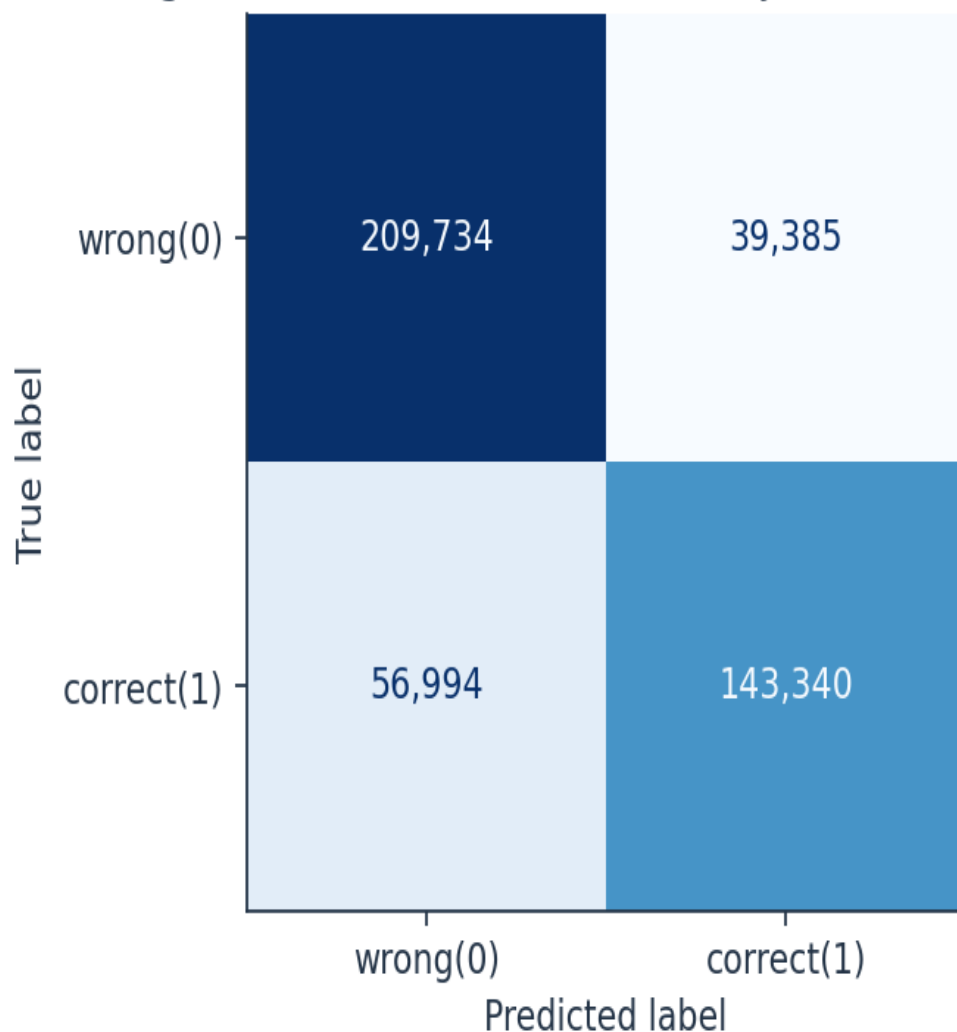


Figure 7c. Confusion matrix for logistic_step_dt at threshold 0.5 on all 449,453 test.json memory (revisit) rows (revisit attempts only; not the superseded first-80k file-order slice).

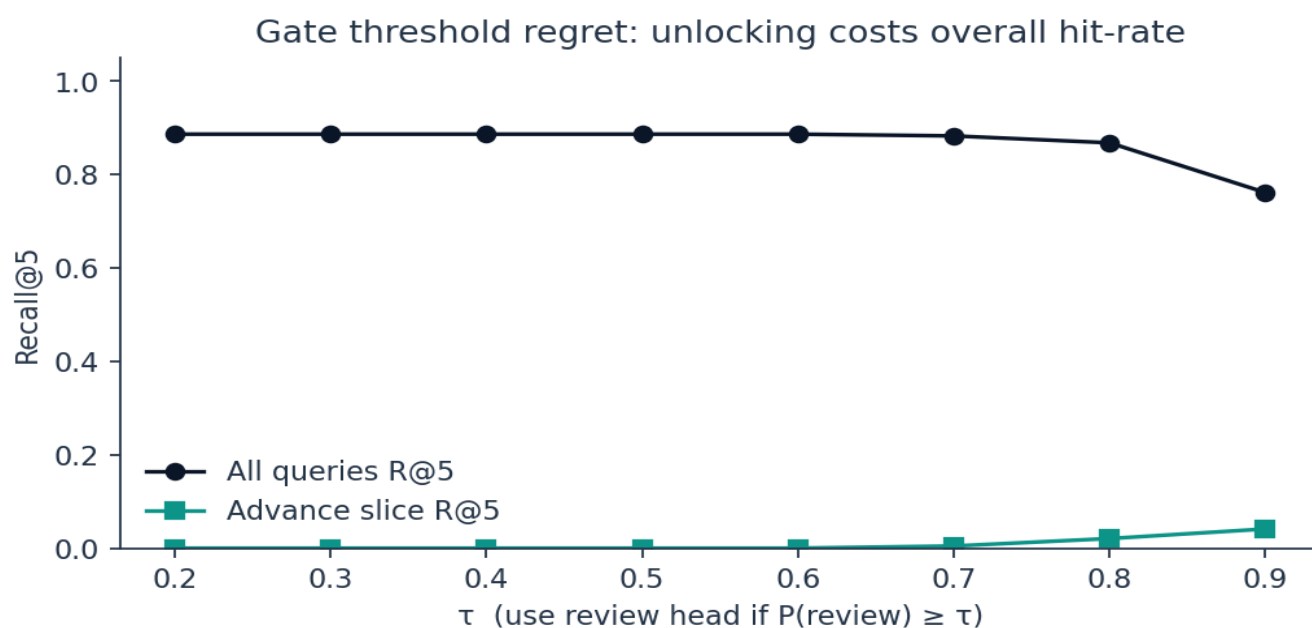


Figure 7d. Raising the review-gate threshold τ forces more advance-head use and trades overall Recall@5 for unlock hits.

6. ROC / AUC (memory classification)

Phase 3 is a binary classifier for **correctness on revisit attempts only**: predict whether the next attempt on a concept already seen in the sequence is correct (label 0/1). The first attempt on a concept emits no row. Models are fit with a **sequence-id split** inside train.json (leakage_check=pass) and scored on official **test.json**.

What the ROC curve represents: trade-off between true-positive rate and false-positive rate as the decision threshold on predicted P(correct) varies. AUC is the probability that a random correct attempt scores higher than a random incorrect one.

Inputs used for ROC/AUC: feature vectors built from prior history in the sequence (success rate, counts, Δt in steps or days / log1p gap); **prediction values** = model's P(correct) scores; **ground truth** = observed correctness of the next attempt.

Model (test.json)	AUC	95% CI	PR-AUC
logistic_step_dt (headline model)	0.864	[0.861, 0.867]	0.826
success_rate	0.858	[0.855, 0.862]	0.815
hlr_step	0.789	[0.784, 0.794]	0.762

Timed learner split (6400 / 1600 users): success_rate AUC 0.697 vs logistic_day 0.694. Paired $\Delta(\text{day-success}) = -0.0029$ [-0.005, -0.001]. forgetting_helps=False on unrestricted timed rows; Follow-up 3 gap $\geq$ 1 redo status=**unresolved**.

ROC / confusion figures in §5 are regenerated from the same protocol (sequence-id fit on train.json → score official test.json).

Gate classifier P(review | history) on Phase 6 queries: AUC 0.749, PR-AUC 0.955, base rate 0.890 (separate binary task: will the true next item be a review vs advance). Full-test calibration (§4.4c): Brier=0.0928, ECE=0.0335, ΔBrier vs base=-0.0085.

7. Complete Dataset → Input → Processing → Output flow

Dataset taken: Junyi Academy / EduData ktbd-junyi (835 concepts, 978 edges, 27,434/6,290 train/test sequences)

↓ Why: only public log with traces + expert DAG in one id space for review/advance recommendation

↓ Phase 1 input: raw ktbd files → filter len 12–200 → output concepts CSV + edges CSV (train interactions 2,342,784; topic n 2,809,159)

↓ Phase 2 input: 835 English names → MiniLM embeddings → concept_embeddings.npy (pair-AUC 0.904; same-topic control 0.464)

↓ Phase 3 input: train sequences (+ timed for day gap) → fit memory classifiers → test.json logistic AUC 0.864; gap $\geq$ 1 forgetting unresolved

↓ Phase 4 input: DAG + prefixes → mix review 0.898, unlock features

↓ Phase 5 input: embeddings + memory + DAG + similarity → gate × (review, advance) heads → serve_models.joblib

↓ Phase 6 (canonical capped): review_ranking.json RAGR R@5 0.891 [0.880, 0.902]

↓ Full-test ranking: paper_ranking_full RAGR R@5 0.889 [0.885, 0.893] (n=6290); advance Markov 0.242 vs RAGR 0.081

↓ Follow-up 3 / paper rank-reversal: concept-level table; exact-dup chance; Markov + GRU/attention probes (not published GRU4Rec/SASRec; RecBole absent)

↓ Probe freeze (val only, seed=20261004): equal-budget GRU+Attn grid (RAGR-lite not in that claim; GRU dropout no-op); test hashed not scored

↓ Prereg draft (§4.13): primary H-rev m $\leq$ 6 Bonferroni; GRU seeds 20261101–03; two-method cluster intersection; freeze-holdout replication after tag + OSF

↓ Final product / diagnostic output: ranked next exercises + P(review) + p_{recall} + subgraph (demo; head softmax calibration deferred; Brier/ECE in §4.4c)

↓ How we show it: figures in §5–6; tables in §4; data cards in docs/data_cards/; JSON under outputs_junyi/phases/; LaTeX under paper_assets/.

Reader map: Dataset + cards — §1 · Phase I/O — §2 · Travel plan — §3 · Actual outputs (freeze §4.11, confirmatory protocol §4.13) — §4 · Charts — §5 · ROC/AUC — §6 · Full trace — §7.

Reproduce: scripts/fetch_datasets.py → run_all_phases.py / run_valid_eval.py → run_followup3.py → run_probe_freeze.py → scripts/run_paper_eval.py → scripts/generate_report_assets.py → scripts/build_paper_assets.py → scripts/build_dataflow_report.py. After prereg-v1 + OSF only: scripts/run_freeze_partition_replication.py --touch-test.